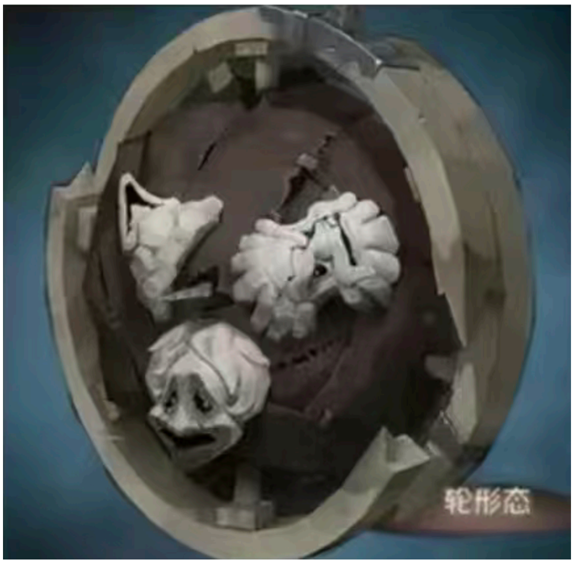
Q **August 3, 2026** Andy He

The Trolley Problem

[6 pts]

Today, SM-senior 黄瑞宣 was kidnapped by the evil Professor 孙笑川. The wicked Emperor has decided to make our great SM-senior 黄瑞宣 solve a problem that has long troubled humanity — and, in doing so, to steal 李文亚's Nobel Prize. After weighing many awards, Professor 孙笑川 concluded that only the Nobel Prize in Physics could suppress 李文亚's white-body biology theory, so he resolved to have 黄瑞宣 solve a physics problem that has baffled all of humankind: the Trolley Problem. 黄瑞宣 had to choose between securing a pentakill for the ace elite and completing an online solo kill. Under Professor 孙笑川's coercion, and with the weight of past memories flooding back, 黄瑞宣 transformed on the spot into the Identity V character 破轮 (whose appearance is shown below).



⋯and crushed Professor 孙笑川 to death, bringing his sinful life to an end.

Treat Professor 孙笑川 as a hemisphere of radius r lying on a plane. 黄瑞宣, having become 破轮, has radius R, mass M, and moment of inertia $I = \frac{1}{2}MR^2$, and has speed v at the instant he crushes into Professor 孙笑川.

Questions.

Q1. Find the kinetic energy lost in rolling over Professor 孙笑川.

Q2. Find the minimum initial speed v for 黄瑞宣 to just barely roll over Professor 孙笑川.